

MOSTAFA ALI MOHAMED ELSHARQAWI

AI / ML Engineer | LLMs · RAG · ComputerVision · MLOps

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PROFESSIONAL SUMMARY

AI/ML Engineer with 2+ years of hands-on experience building and deploying end-to-end AI systems. Delivered an **A+-graded AI recruitment platform (Hire-Mind)** leveraging RAG, semantic search, and LLM-powered interview coaching — deployed via Docker on a scalable FastAPI + MongoDB stack. Built **SmartDetect**, a real-time Computer Vision system using YOLOv8 + EfficientNet achieving sub-100 ms inference. Proficient in the full AI lifecycle: data preprocessing, model training (PyTorch / TensorFlow), evaluation, optimization, and production deployment on AWS and Google Cloud (Vertex AI). Experienced with Autonomous Agents (CrewAI), LangChain, Prompt Engineering, and MLOps best practices.

TECHNICAL SKILLS

AI & LLMs	LLMs, Generative AI, RAG, Retrieval-Augmented Generation, LangChain, CrewAI, Autonomous Agents, Prompt Engineering
Deep Learning	PyTorch, TensorFlow, Transformers, Hugging Face, YOLOv8, EfficientNet, CNNs, Neural Networks, Fine-tuning
NLP	Text Embeddings, Semantic Search, Cosine Similarity, Tokenization, Named Entity Recognition, Sentence Transformers
Computer Vision	Object Detection, Image Classification, Quality Assessment, OpenCV, PyQt, Real-time Inference, Multi-stage Pipelines
AI APIs & Models	OpenAI API, Anthropic Claude, Google Gemini, Mistral, LLaMA, Hugging Face Hub, MLflow
Backend & APIs	FastAPI, Django, REST APIs, Async Python, WebSockets
MLOps & Deploy	Docker, CI/CD, MLflow, RunPod, Vast.ai, Model Optimization, Inference Acceleration
Cloud	AWS (EC2, S3, Lambda), Google Cloud Platform, Vertex AI
Databases	PostgreSQL, MongoDB, Redis, ChromaDB (Vector DB)
Data Engineering	Scikit-learn, Pandas, NumPy, Data Preprocessing, Feature Engineering, Evaluation Metrics (mAP, F1, AUC)

PROJECTS

Hire-Mind — AI-Powered Recruitment System

2024 – 2025

- GraduationProject | Grade:A+ | 2024–2025
- Architected an end-to-end AI recruitment platform using **NLP, RAG, and semantic search** to match candidate CVs with job descriptions; system handles multi-step retrieval with re-ranking.
 - Engineered a **FastAPI** backend with sentence-transformer embeddings + **Cosine Similarity** scoring, persisted in **MongoDB**; API serves results in <200 ms.
 - Built an **LLM-powered AI interview trainer** (LangChain + OpenAI) that generates role-specific questions and real-time feedback for candidates.
 - Containerized all services with **Docker Compose**; modular microservice architecture allows independent horizontal scaling of each component.
 - Achieved highest academic grade (A+); selected as a showcase project for the university AI department.

<https://github.com/mstfyshrqawy520-alt/HireMind-AI-Recruitment-System>

NTIScholarship | 120TrainingHours | 2024

- Designed a **multi-stage CV pipeline**: fruit detection (YOLOv8), classification (EfficientNet-B4 / PyTorch), quality grading, and size estimation — all in a single inference pass.
- Achieved **sub-100 ms end-to-end inference** on a FastAPI backend with confidence-based validation to suppress false positives across all pipeline stages.
- Implemented **multi-class confidence thresholding** with fallback logic, reducing false-positive rate by ~30% compared to single-threshold baseline.
- Delivered a **PyQt5 + OpenCV** desktop GUI with real-time camera feed, bounding-box overlays, class labels, and export to CSV report.
- Full MLOps setup: model versioning with **MLflow**, Dockerized inference server, deployable on RunPod GPU cloud.

<https://github.com/mstfyshrqawy520-alt/SmartDetect-fruite>

EXPERIENCE

IEEE Volunteer – AI & ML Team Member2025 – Present

IEEE Beni Suef Student Branch, Egypt

- Organized and delivered AI/ML workshops attended by 100+ engineering students, covering topics from supervised learning to LLM applications.
- Led hands-on sessions on Generative AI tools and Prompt Engineering as part of the branch's technical outreach program.
- Collaborated with a cross-functional team to produce technical content and coordinate event logistics.

EDUCATION

B.Sc. in Computer Science — Artificial Intelligence SpecializationExpected 2026

Egyptiane-LearningUniversity(EELU), Egypt

- Relevant coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Data Structures & Algorithms, Database Systems.

CERTIFICATIONS

NVIDIA DLI	Deep Learning (90 hrs) + Building LLM Applications with Prompt Engineering
DEPI / MCIT	Digital Egypt Pioneers Initiative — Generative AI Track (6 Months)
NTI / ITIDA	Machine Learning Track (120 hrs)
ITI	Artificial Intelligence & Machine Learning (72 hrs)
DataCamp	Machine Learning Scientist in Python Deep Learning NLP Computer Vision
Sprints x Microsoft	AI & Machine Learning Summer Camp (40 hrs) — Credential: SPR-TN945C
Cisco NetAcad	Introduction to AI Switching, Routing & Wireless Essentials

LANGUAGES

Arabic — Native ProficiencyEnglish — B2 Upper-Intermediate (Professional Working Proficiency)